

Sales Forecasting and Inventory Efficiency at Riddhi Siddhi Medical Store

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1.Executive Summary:-

This project, titled “Optimization of Sales and Inventory Management at Riddhi Siddhi Medical Store,” addresses two pressing challenges faced by the pharmacy: declining sales and inefficient inventory control. The business operates in a competitive local market with over 25 medical shops and around 10 hospitals and struggles with stock expiry, overstocking, understocking, and inconsistent demand forecasting. To tackle these issues, a multi-pronged analytical approach was adopted, combining descriptive, predictive, and prescriptive analytics using sales and inventory data from November 2024 to January 2025.

The analysis yielded several critical findings. An ABC classification revealed that while generic medicines constitute over 84.7% of inventory value, value is concentrated at the product level, not the category level, demanding more granular monitoring. This need for deeper analysis was reinforced by the discovery that 17% of stock expired unsold, pointing to gaps in inventory rotation. Further investigation into overstocking and understocking uncovered significant mismatches, such as excess stock of slow-movers like Metol X-1.25 and shortages of highdemand products like Paracetamol. To identify top-performing products, a Data Envelopment Analysis (DEA) highlighted high-efficiency medicines like Saroglitazar, which provide excellent returns, in contrast to underperformers like Rexcof. To forecast future demand, a predictive XGBoost model was developed that accurately predicts daily sales trends with an R^2 of 0.7868. Finally, RFM segmentation confirmed that a small subset of medicines drives the majority of revenue, reinforcing the need for focused stocking strategies on these key items.

Based on these data-driven insights, this report recommends a framework to boost sales and optimize inventory. The pharmacy should improve inventory efficiency by using DEA and RFM segmentation to prioritize high-performing medicines and phase out or discount nonperforming stock. To prevent financial losses, the implementation of real-time stock monitoring and automated expiry alerts is crucial. Furthermore, sales can be enhanced by leveraging the predictive model for demand forecasting and implementing product-level ABC controls.

By integrating these practices into its daily operations, Riddhi Siddhi Medical Store can significantly reduce operational costs, prevent waste, and enhance revenue performance. The outcome is a sustainable, analytics-enabled decision-making framework tailored to the real challenges of a modern retail pharmacy.

Scope and Limitations

This project performs an analysis of sales and inventory based on a three-month dataset from November 2024 to January 2025. It is important to acknowledge the limitations of this timeframe. While the data provides valuable initial insights into product performance, shortterm demand imbalances, and operational efficiency, it is insufficient to capture long-term trends or seasonal demand fluctuations inherent in the pharmaceutical retail cycle.

2.Detailed Explanation of Analysis Process/Method:-

2.1 Data Collection and Initial Preprocessing

The data for this project was sourced from the Riddhi Siddhi Medical Store's internal records, covering sales and inventory from November 2024 to January 2025. The raw data was provided in two separate CSV files: one for sales transactions and one for inventory logs. Before any specific analysis, a global data preprocessing routine was applied. This involved:

- **Standardizing Column Names:** All column names were converted to a consistent format (e.g., lowercase with underscores) to prevent errors during data merging.
- **Data Type Conversion:** Date columns, such as 'sale_date' and 'expiry_date', were converted to a standard datetime format to enable time-based calculations. Numeric columns like 'quantity' and 'price' were checked and converted to appropriate float or integer types.
- **Handling Inconsistencies:** The 'Medicine Name' column was cleaned to resolve inconsistencies (e.g., 'Paracetamol 500mg' vs. 'Paracetamol-500') to ensure accurate merging between the sales and inventory datasets.

2.2 ABC Classification of Medicine Inventory

ABC Classification of Medicine Inventory

ABC analysis is a method of inventory categorization based on the Pareto principle, which states that a small percentage of items account for the largest share of value. The process

involved merging the sales and inventory datasets and calculating the inventory value for each medicine category.

Inventory Value Formula: The monetary value for each item was calculated to understand its contribution.

$$\text{Inventory Value} = (\text{Closing Stock} + \text{New Stock}) \times \text{Selling Price}$$

This formula represents the total retail value of all stock available for each medicine during the period, providing a clear basis for value-based categorization.

Classification Formula: Items were sorted by descending inventory value. The cumulative percentage of total value was then calculated to classify items into categories:

$$\text{Cumulative \%} = \sum_{k=1}^{\sum_{j=1}^n \text{Value}_j} \frac{\text{Value}_k}{\sum_{j=1}^n \text{Value}_j} \times 100$$

Based on this classification method, the inventory items were segmented into three categories. Category A represents the most critical items, accounting for the top 70% of the total inventory value. Category B includes items of moderate importance, which make up the next 20% of the value, while Category C consists of the least valuable items, comprising the final 10%.

2.3 Expired Stock Utilization and Waste Analysis :-

Data Preparation and Identification of Expired Stock

To accurately identify expired medicines, the initial step involved data standardization. All column names were normalized for consistency, and the expiry date field was converted into a standard datetime format for reliable comparison. Using this prepared data, any medicines with expiry dates earlier than the current date were flagged as "expired inventory," creating a clear dataset of all obsolete stock.

Analysis of Stock Utilization

The list of expired inventory was then merged with the complete sales dataset. The primary goal of this step was to evaluate whether these medicines were sold before their official expiry date.

This matching process enabled a direct measurement of how efficiently the store's stock was being utilized before it reached the point of becoming waste.

Measurement of Waste and Inefficiency

From this analysis, three key performance metrics were computed: the total units sold before expiry, the total number of expired units, and the specific count of units that expired without ever being sold. These final metrics are crucial as they directly highlight inefficiencies in the pharmacy's current demand forecasting and inventory rotation practices.

2.4 Overstocking vs Understocking Trend Analysis

Dataset Preparation

All date fields were converted into datetime format to ensure reliable grouping. Stock and sales data were aggregated by medicine name to produce a consolidated view.

Surplus Calculation

A new metric, surplus, was calculated as:

$$\text{Surplus} = \text{Total New Stock} - \text{Total Sold Quantity}$$

Positive surplus indicated overstocking, while negative surplus signaled understocking.

Visualization and Insight Generation

Two visualizations were created: one showing surplus values across all medicines, and another zoomed in on the top 10 overstocked and understocked medicines. These visual tools were instrumental in identifying key mismatches and optimizing reorder policies.

2.5 DEA-Based Efficiency Assessment of Medicines

DEA Model Setup

Data Envelopment Analysis was used to evaluate how efficiently each medicine converts stock into profit. Each medicine was treated as a Decision-Making Unit (DMU).

Inputs and Output

To measure the efficiency of each medicine, the Data Envelopment Analysis (DEA) model was structured with specific inputs and one output. The model utilized two key inputs: the total units sold, which acts as a proxy for demand, and the total new stock, representing the inventory investment. The single output was the total profit earned from each medicine, allowing the model to evaluate how efficiently inputs were converted into financial return.

DEA Model (Input-Oriented): The efficiency (theta) is found by solving:

$$\min \theta$$

$$\min \theta$$

$$\min \theta \text{ s.t. } \sum_{j=1} \lambda_j x_{ij} \leq \theta x_{io}, \sum_{j=1} \lambda_j y_{rj} \geq y_{ro}, \lambda_j \geq 0$$

Linear Programming Optimization

The DEA problem was formulated and solved using PuLP. Constraints ensured no medicine appeared more efficient than another under the same weight structure. Efficiency scores were normalized to lie between 0 and 1, where 1 indicates full efficiency.

This method highlighted medicines that deliver optimal financial return per stock unit, as well as those requiring pricing or stocking adjustments.

2.6 Predictive Sales Modeling with XGBoost

Model Rationale and Selection

In the midterm analysis, both **Linear Regression** and **Seasonal ARIMA (SARIMA)** models were explored for sales forecasting. However, the Linear Regression model exhibited signs of **overfitting**, capturing noise rather than underlying trends, while SARIMA **underfit** the data and lacked flexibility to incorporate external variables such as product features or inventory behavior.

To address these limitations, the final phase employed an **XGBoost regression model**, known for its robustness with non-linear data and ability to model complex feature interactions. A **hyperparameter tuning strategy** was adopted to optimize model performance, improving generalization and reducing error rates compared to earlier models.

Data Preprocessing and Feature Engineering

Outlier Handling:

Boxplots were used to detect extreme values in numeric fields such as stock levels and sales quantities. Outliers were capped or excluded using **interquartile range (IQR)** thresholds to prevent skewed predictions.

Date Feature Transformation:

The date column was decomposed into separate features: Year, Month, Day, and DayOfWeek to allow the model to learn time-based sales patterns. The original date column was then dropped to avoid redundancy.

Encoding and Transformation:

Categorical features were encoded using **OneHotEncoder**, while numeric features were passed through unchanged. These transformations were applied using a

ColumnTransformer within a **scikit-learn pipeline** for seamless preprocessing and model training.

Model Training and Hyperparameter Tuning

The dataset was split into **training and test sets (80:20)** to evaluate out-of-sample performance. The preprocessing steps and XGBoost model were combined into a single pipeline.

To optimize performance, key hyperparameters such as:

`max_depth`, `learning_rate`, `n_estimators`, `subsample`, `colsample_bytree`

were tuned based on prior exploratory runs. The final model used the following best settings:

```
XGBRegressor( objective='reg:squarederror', colsample_bytree=0.8,
learning_rate=0.1, max_depth=5, n_estimators=100, subsample=0.8,
random_state=42 )
```

The trained model was integrated into the pipeline and fitted on the training set. Its predictions were then compared to actual values from the test set.

Model Evaluation Metrics

The model's predictive performance was evaluated using standard regression metrics:

Mean Absolute Error (MAE): 15.23

Using this formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|$$

Root Mean Squared Error (RMSE): 21.42

Using this formula:

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

R² Score (Coefficient of Determination): 0.7868

Using this formula:

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}$$

These results demonstrate a strong model fit, capturing nearly 79% of the variance in daily sales. The low residual spread and close alignment between predicted and actual

sales values make the model well-suited for deployment in automated sales forecasting systems.

2.7 RFM Segmentation of Medicine Sales

RFM Feature Engineering

To evaluate medicine performance based on sales behavior, RFM (Recency, Frequency, Monetary) segmentation was applied. The sales dataset was grouped by medicine name, and three key metrics were computed.

Recency was calculated as the number of days since a medicine's last recorded sale.

Frequency was the total number of distinct sale events for each medicine , and

Monetary was the cumulative revenue generated from each medicine's total sales amount. To enable effective comparison across these different dimensions, each metric was then normalized using MinMaxScaler, scaling all values to a consistent range between 0 and 1. This process creates a standardized behavioral summary of each product's market activity.

Bubble Chart Analysis and Segmentation

The primary analysis was conducted using a bubble chart designed to capture all three RFM dimensions simultaneously. In this visualization, the x-axis represents Recency, the y-axis represents Frequency, and the bubble size corresponds to the Monetary value. The chart clearly reveals that top-performing products cluster in the bottom-right quadrant, which represents medicines sold both frequently (high frequency) and recently (low recency). These products also have a high revenue impact, indicated by their large bubble size. Conversely, medicines with high recency and low frequency, represented by small bubbles in the upperleft quadrant, can be identified as poor performers or outdated stock.

Strategic Applications of RFM Analysis

The insights from this segmentation support several data-driven strategies for inventory and promotion. Key actions include focusing on retaining high-value products through consistent restocking , phasing out slow-moving medicines to free up capital and shelf space , and designing targeted promotional campaigns for underutilized items that have the potential to become more profitable.

3.Results and Findings :-

3.1 ABC Classification of Medicine Inventory

Visual Analysis

The bar chart below illustrates the distribution of total inventory value across three primary medicine categories: Generic, Branded, and Ayurvedic. The height of each bar represents the total value (in Indian Rupees) for each category. Categories are colorcoded by ABC classification: orange for Category B and green for Category C. Notably, no category was classified as 'A' at the category level, suggesting that high-value items may be concentrated at the individual product level within the Generic category.

Observations and Interpretation

As seen in Figure 1, the Generic category holds the highest inventory value, contributing over ₹2.8 million, which accounts for approximately 84.7% of the total value. Despite this dominance, it is classified as Category B due to the cumulative method of ABC classification at the category level. The Branded and Ayurvedic categories contribute ₹317,550 and ₹190,160 respectively, making up the remaining 15% and falling under Category C.

This uneven distribution of inventory value supports the Pareto principle, where a small portion of stock contributes to a large share of value. The implication is that the Generic category deserves focused attention in terms of procurement planning, expiry management, and sales optimization.

These results provide a clear basis for targeted stock monitoring, emphasizing that a more granular product-level ABC analysis might be necessary to isolate true Category A items.

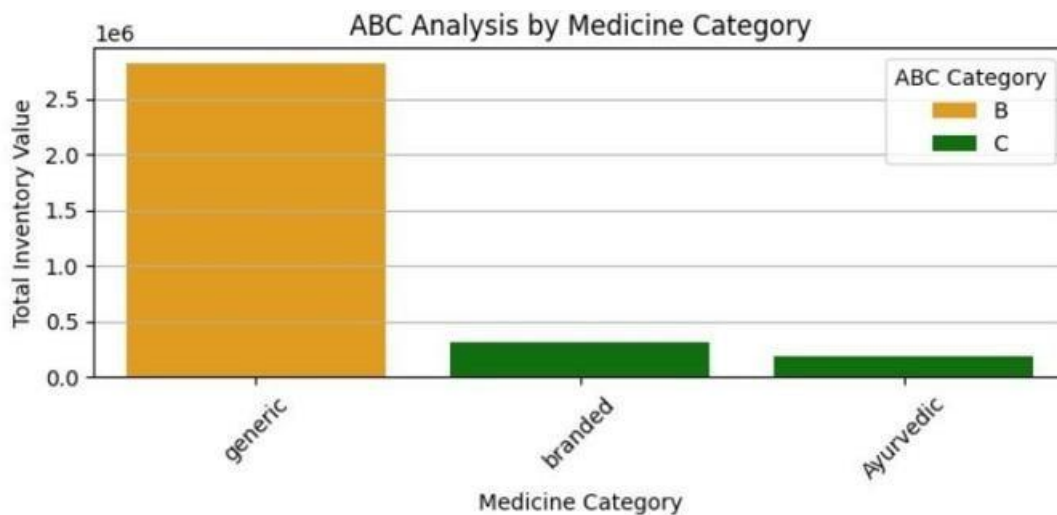


Figure 1: ABC Analysis by Medicine Category

3.2 Expired Stock Utilization and Waste Analysis

Visual Analysis

Figure 2 illustrates the utilization status of expired stock using two visualizations: a bar chart (left) showing the number of units sold before expiry versus those that expired unsold, and a pie chart (right) displaying the proportional distribution of the same.

Observations and Interpretation

The bar chart indicates that approximately 220 units were sold before expiry, whereas around 45 units expired without being sold. This data is further confirmed by the pie chart, which shows that 83% of the expired stock was utilized effectively, while 17% resulted in wastage.

These results suggest that while the store has a relatively effective stock utilization strategy, the 17% unsold expired inventory represents a tangible financial and operational inefficiency. Products with short shelf life or low demand should be reviewed, and stock rotation practices such as First-ExpiryFirst-Out (FEFO) should be reinforced.

Automated expiry alerts, improved demand forecasting, and tighter control on overstocking could further reduce the proportion of unsold expired stock.

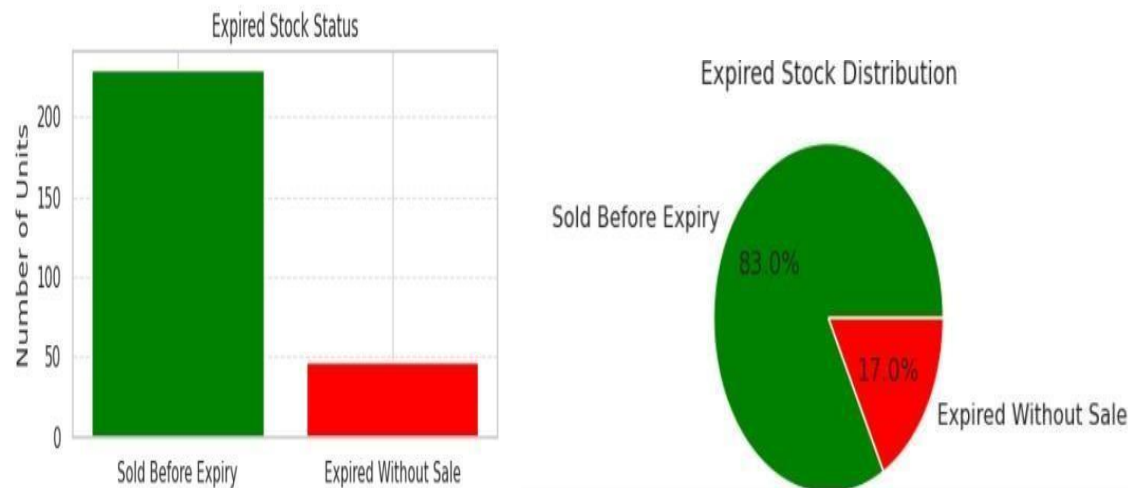


Figure 2: Expired Stock Utilization and Waste Analysis

3.3 Overstocking vs Understocking Trend Analysis

Visual Analysis

The bar chart in Figure 3 displays the surplus values for different medicines, where surplus is defined as the difference between new stock and sold quantity. Medicines with positive surplus values are shown in green and represent overstocked items, while those with negative values are displayed in red and indicate understocking.

Observations and Interpretation

It is evident from the chart that medicines such as Metol X-1.25, Renitum Gold, and Cheston Cold have significantly high surplus values, suggesting that they were overstocked beyond actual sales requirements. In contrast, items such as Ofokem-200, Paracetamol, and Dynaflex Spray exhibit negative surplus values, implying that demand exceeded supply, leading to potential stockouts.

Identifying such inventory mismatches is essential for improving stock planning and replenishment strategies. Overstocking leads to capital lock-in and increased expiry risk, while understocking can result in lost sales and dissatisfied customers. Regular analysis using this method allows the store to fine-tune its inventory policies based on real sales trends.

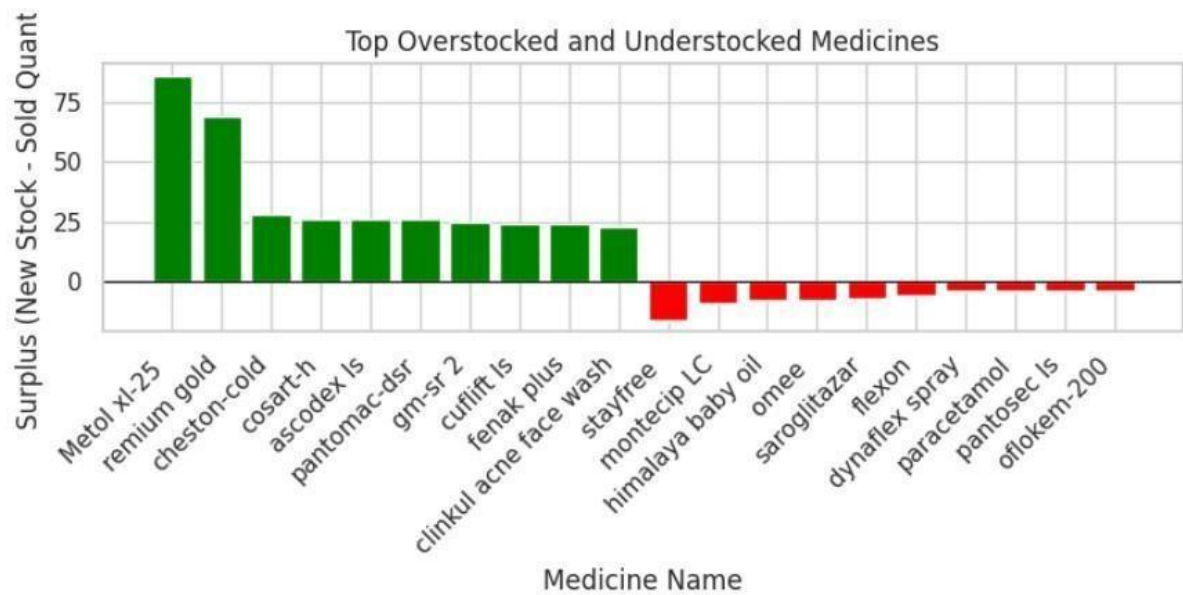


Figure 3: Top Overstocked and Understocked Medicines

3.4 DEA-Based Efficiency Assessment of Medicines

Visual Analysis

Figure 4 and **Figure 5** presents two plots related to DEA efficiency scores. The first chart shows the distribution of efficiency scores across all medicines, segmented into intervals. The second chart ranks the top 20 medicines based on their DEA scores, with values ranging from 0 to 1.

Observations and Interpretation

The upper chart reveals that the majority of medicines have DEA efficiency scores between 0.0 and 0.4, suggesting substantial room for improvement. A small fraction of products demonstrate high efficiency, indicating that they offer better profit returns for the stock invested.

The lower chart highlights the top 20 most efficient medicines. Medicines such as Saroglitazar, Swarna Bhasma, and Liv 52 show scores approaching or equal to 1.0, meaning they are optimal in terms of sales-driven profitability and inventory management. On the other hand, commonly stocked items such as Rexcof and Cuffit LS scored lower, indicating potential overstocking or poor turnover.

These insights guide procurement strategies, suggesting which products should be prioritized for stocking and which may need discontinuation, discounting, or promotional support to improve turnover.

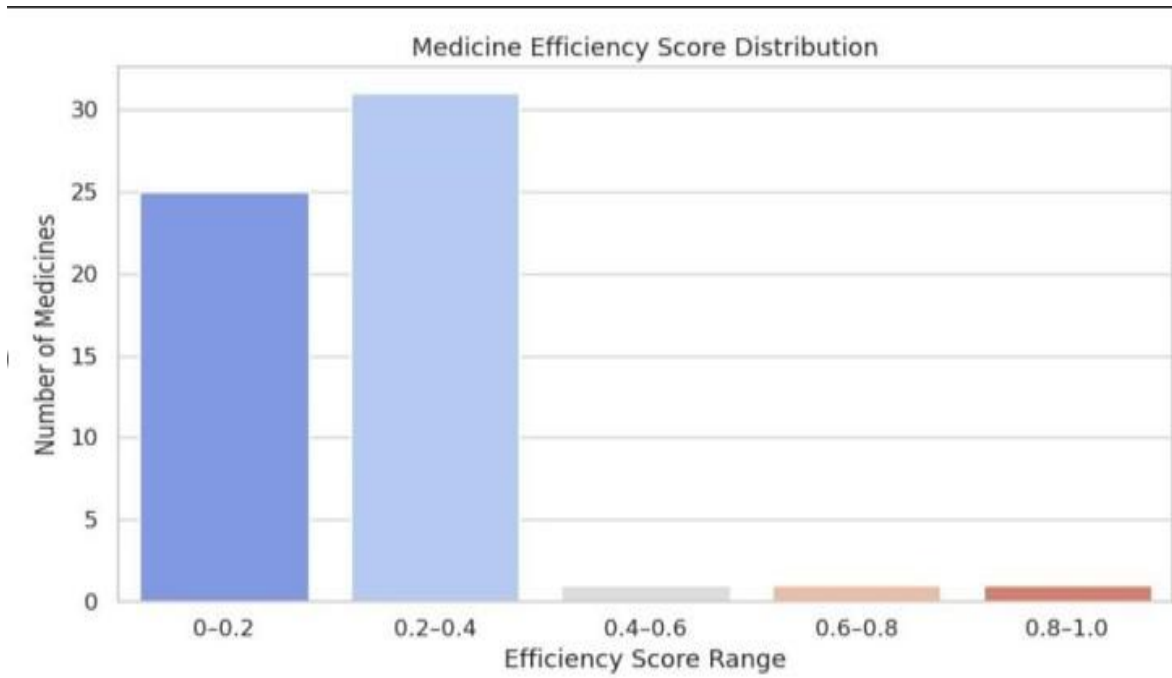


Figure 4: Distribution of DEA Scores

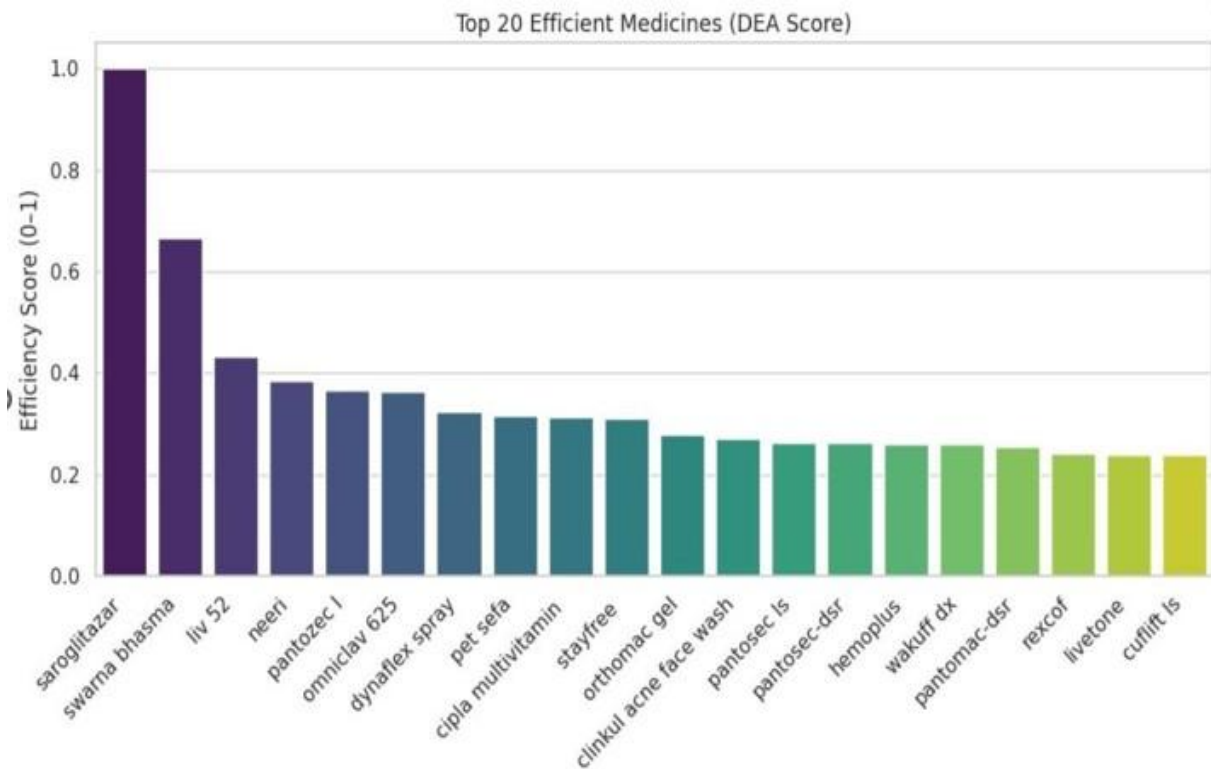


Figure 5: Top 20 Efficient Medicines

3.5 Predictive Sales Modelling with XGBoost:-

Figure 6 presents a visual time series comparing the actual daily sales to the sales predicted by the model from November 2024 to February 2025. The predicted sales line closely tracks the actual sales trend, demonstrating a strong temporal model fit and the model's ability to capture daily fluctuations, including major sales peaks. This strong visual evidence is supported by the model's R^2 score of 0.7868, which indicates that it explains approximately 79% of the variance in daily sales. To further validate the model for systemic bias,

Figure 7 displays a histogram of the model's residuals, or prediction errors. The distribution is centered around zero and shows a near-normal spread, confirming that the model's errors are consistent and not skewed towards systematic over- or under-prediction. Taken together, these figures demonstrate that the XGBoost model is both accurate in its sales forecasting and statistically reliable.

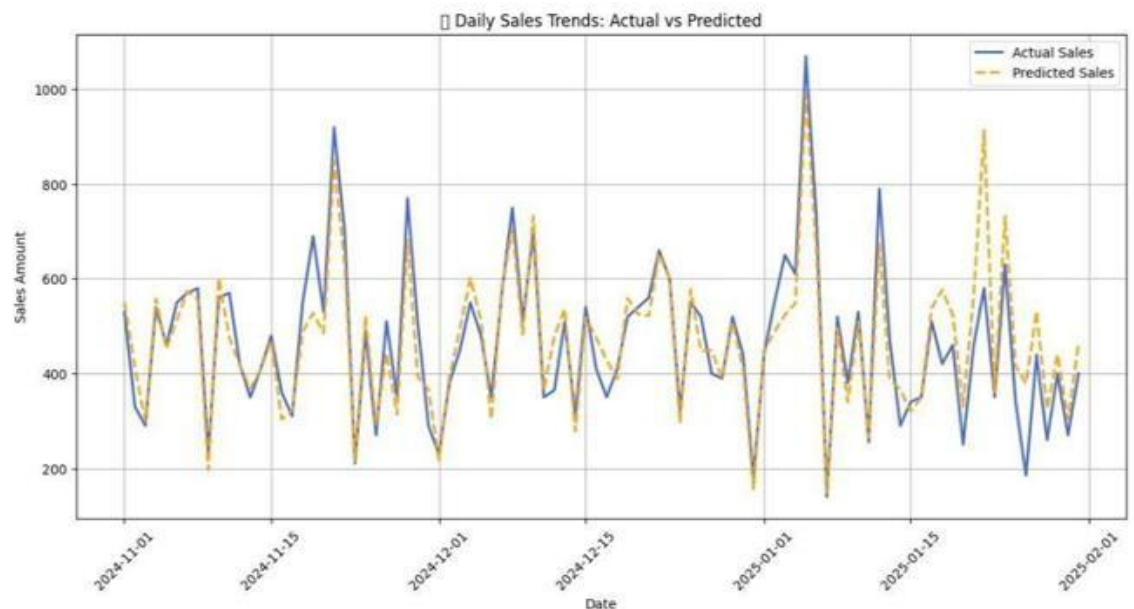


Figure 6: Daily Sales Trends - Actual vs. Predicted

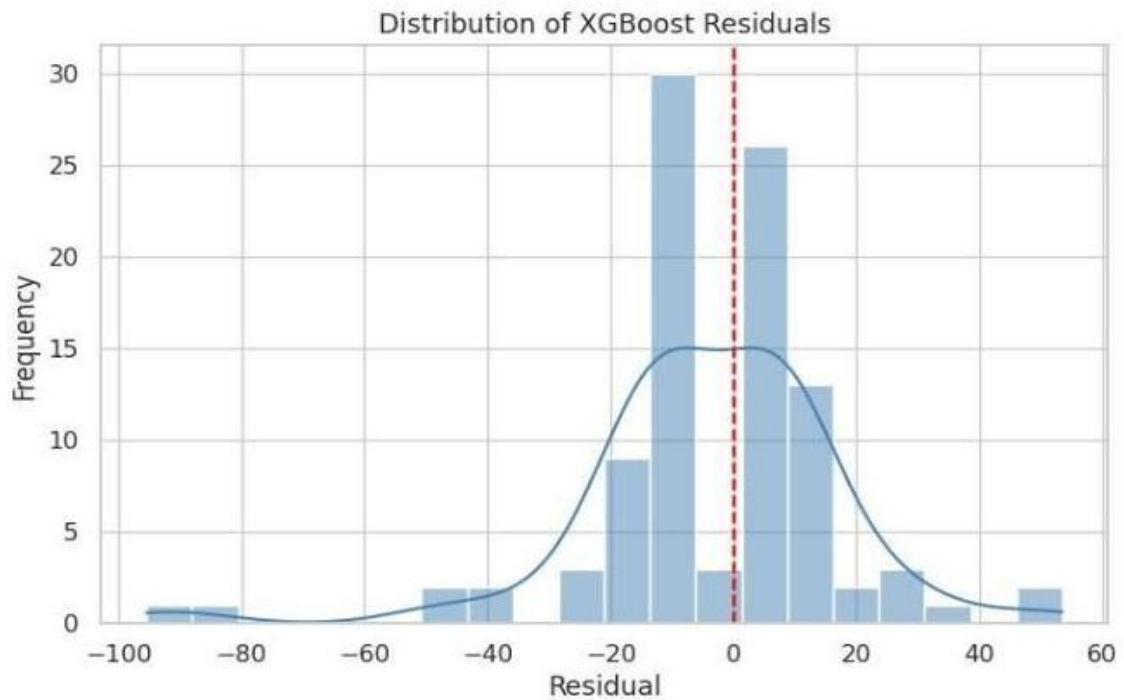


Figure 7: Distribution of XGBoost Residuals

3.6 RFM Segmentation of Medicine Sales:-

Figure 8, which plots three key performance dimensions simultaneously to identify top products. The chart's x-axis represents Recency (fewer days since the last sale is better), the yaxis shows Frequency (a higher number of sales is better), and the size of each bubble corresponds to its total Monetary value. Medicines clustered in the bottom-right quadrant (low recency, high frequency) represent the most active and valuable products. The largest bubbles in this area are the store's top performers, driving significant and consistent revenue. Conversely, the small bubbles in the top-left quadrant (high recency, low frequency) signify potentially underperforming or outdated stock that contributes little to sales. This single chart provides a clear guide for inventory strategy, distinguishing high-priority medicines that must be kept in stock from low-priority items that may require promotional discounts or discontinuation.

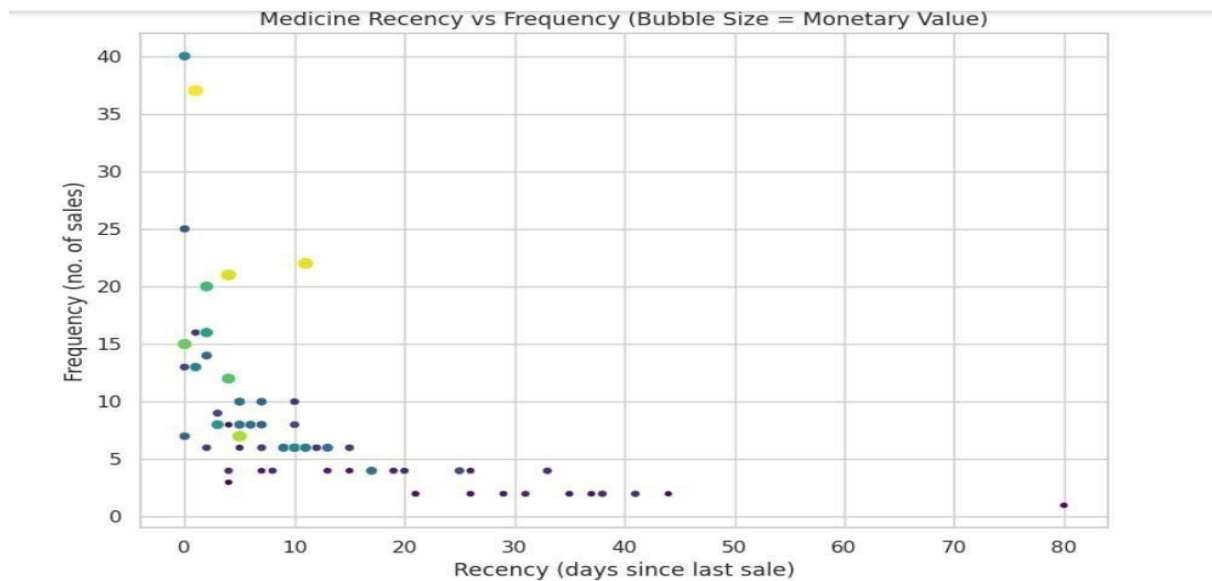


Figure 8: RFM Segmentation of Medicines

4. Interpretation of Results and Recommendation:-

4.1 Tackling Declining Sales

4.1.1 Understanding Product Contributions with ABC Analysis

The ABC classification revealed that a substantial share of inventory value (84.7%) resides in generic medicines (**Figure 1**), yet no category qualified as 'A' at the grouped category level. This suggests that the business's most valuable products are spread across individual medicines rather than entire categories.

Implication: The store lacks targeted focus on high-value individual medicines.

Recommendation:

Implement a **product-level ABC classification** to pinpoint true "A" items and apply focused promotional, pricing, and replenishment strategies on them. For example, highlight top A-medicines in customer-facing promotions and ensure they remain in stock.

4.1.2 Forecasting Sales Trends with XGBoost:-

The XGBoost model, trained on historical sales and medicine features, achieved an R^2 of 0.7868 (**Figure 6**), indicating it can explain about 79% of the variance in sales. Residuals followed a near-normal distribution, confirming stable predictive performance.

Implication: Predictive modeling can play a pivotal role in combating declining sales by enabling better demand planning and pricing.

Recommendation:

Deploy this model within a sales dashboard to **generate daily sales forecasts**. Integrate forecasts into **procurement and promotion cycles**, helping avoid missed sales due to understocking and identifying when to push specific products with expected demand spikes. Regularly retrain the model quarterly with fresh data.

4.1.3 Segmenting Medicines with RFM Analysis

RFM segmentation (**Figure 8**) revealed distinct performance patterns across products. Medicines in the low-recency, high-frequency, and high-monetary segment are the true sales drivers.

Others showed high recency and low frequency, indicating potential obsolescence.

Implication: The store treats all products equally despite their differing sales behaviors and revenue contributions.

Recommendation:

Adopt **tiered product strategies**:

For Tier 1, top-performing medicines should be pushed with regular restocking and bundling offers. Tier 2 should consist of moderately selling medicines that are consistently monitored and reassessed. For Tier 3, the store should consider phasing out poorly performing medicines or use discounts to clear this stagnant stock.

4.2 Optimizing Inventory Management

4.2.1 Reducing Expiry Losses with Expiry Waste Analysis

The analysis revealed that 17% of expired medicines remained unsold (**Figure 2**), pointing to a tangible financial drain. Despite good overall usage of stock, medicines with short shelf lives or weak demand still end up wasted.

Implication: Forecasting isn't adequately accounting for shelf life and slow turnover.

Recommendation:

Implement **FEFO (First-Expiry-First-Out)** logic in inventory management and issue **expiry alerts** for medicines nearing the expiry window. Adjust reorder quantities based on demand forecasting combined with shelf-life risk. Consider markdown pricing for medicines within 30 days of expiry.

4.2.2 Balancing Inventory with Overstocking/Understocking Analysis

Some medicines like Metol X-1.25 and Cheston Cold were significantly overstocked (**Figure 3**), while fast-moving items like Paracetamol were understocked. This mismatch reflects inconsistent forecasting and reorder policies.

Implication: Overstocking leads to capital lock-in and expiry risks; understocking leads to lost sales.

Recommendation:

Integrate **dynamic inventory dashboards** to flag unusual surplus/deficit patterns in realtime. Review demand forecasts weekly and adjust purchase plans accordingly. For surplus stock, offer **bulk discounts** or **bundle promotions** to accelerate turnover.

4.2.3 Prioritizing Efficient Products with DEA

DEA analysis revealed that most medicines had efficiency scores between 0.0 and 0.4 (**Figure 4**), meaning that most do not convert investment into profit effectively. A few, such as Saroglitazar and Swarna Bhasma, scored near 1.0, indicating ideal performance.

Implication: Inventory investment is not being optimized around the most profitable and efficient products.

Recommendation:

It is recommended to use DEA efficiency scores as a core decision-making metric during purchase planning. This involves continuing to invest in high-efficiency products while taking corrective action for low-efficiency items by re-evaluating their pricing, negotiating better purchase rates, or reducing their procurement volume. To ensure this is an ongoing practice, the DEA evaluations should be automated using existing sales and inventory data at the end of each quarter.